

CGN 3405: Applied Numerical Methods for Civil Engineering

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Class Room: NSC O110; Nicholson School of Communication and Media

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1 Points

[Important] Make sure you complete each Quiz session before the given due time. If you missed the Quiz, please reach out immediately. Late completion would have 10% penalty.

Table 1: Points of class participation, Quizzes, assignments, and exams

Items	Class	Quizzes	Assignment	Exam 1	Exam 2	Final Exam
Points	100	300	400	300	300	600
Portion	5%	15%	20%	15%	15%	30%
Amount	-	300	7	-	-	-

1.1 Quizzes Questions (300 points in total)

- Week 1: $5 + 12 + 14 = 31$
- Week 2: $15 + 11 = 26$
- Week 3: $13 + 14 + 9 = 36$
- Week 4: $10 + 14 + 13 = 37$
- Week 5: $10 + 10 = 20$
- Week 7: $10 + 9 + 8 = 27$
- Week 8: $10 + 7 + 14 = 31$
- Week 9: $10 + 7 + 8 = 25$
- Week 10: $9 + 10 + 5 = 24$
- Week 12: **17**
- Week 13: **26**

Progress: **300/300** (Updated on April 22, 2026)

1.2 Assignment (400 points in total)

Table 2: Points of 7 assignments

Items	A1	A2	A3	A4	A5	A6	A7
Points	75	50	55	55	55	55	55

Progress: **400/400** (Updated on April 13, 2026)

2 Teaching Schedule

2.1 Schedule

Please check out **14-week teaching schedule** in Table 3.

2.2 Assignments

Please check out **7 assignments** in Table 3.

2.3 Exams

Please check out **2 written middle-term exams** and the **1 final written exam** in Table [3](#).

- **Exam 1** (written test, in-class): **2:30PM – 3:20PM, February 20, 2026**
 - 20 multiple choice questions (40%)
 - 4 computing questions (40%)
 - 3 Python programming questions (20%)
- **Exam 2** (written test, in-class): **2:30PM – 3:20PM, April 10, 2026**
 - 20 multiple choice questions (40%)
 - 5 computing questions (60%)
- **Final Exam** (written test, in-class): **1:00PM – 3:50PM, April 29, 2026**
 - 9 computing questions (88%)
 - 3 programming questions (12%)

Table 3: Teaching schedule of CGN 3405 (Spring 2026)

Week	Teaching Content	Date	Note
1	Introduction to the Course & Logistics	01/12 01/14 01/16	
2	Mathematical Modeling & Engineering Problem Solving	01/19 01/21 01/23	Martin Luther King Jr. Day Release Assignment 1
3	Introduction to Python Programming: Part I	01/26 01/28 01/30	
4	Introduction to Python Programming: Part II	02/02 02/04 02/06	Release Assignment 2
5	Modeling and Errors: Part I	02/09 02/11 02/13	
6	Modeling and Errors: Part II	02/16 02/18 02/20	Release Assignment 3 Review Class △△ Exam 1
7	Nonlinear Equations	02/23 02/25 02/27	
8	Introduction to Applied Linear Algebra: Part I	03/02 03/04 03/06	Release Assignment 4
9	Introduction to Applied Linear Algebra: Part II	03/09 03/11 03/13	Release Assignment 5
10	Linear Algebraic Equations	03/23 03/25 03/27	Release Assignment 6
11	Ordinary Differential Equations	03/30 04/01 04/03	Review Class
12	Optimization Techniques: Part I	04/06 04/08 04/10	△△ Exam 2
13	Optimization Techniques: Part II	04/13 04/15 04/17	Release Assignment 7
14	Curve Fitting	04/20 04/22 04/24	Review Class Review Class
15	Final Exam	04/27 04/29	No class, self study △△ Final Exam